

CSE 423 Computer Graphics

Quiz -3 (20 marks)

Duration: 20 minutes

Name:	ID:	Section: 10
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- Given a point P (10, 20, 40), calculate the projected point P' using Cavalier Projection, orientation angle is 30° and projection plane Z = 5

8

$$P' = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{[Move plane back to } Z=5\text{]}} \cdot \underbrace{\begin{bmatrix} 1 & 0 & 0.866 & 0 \\ 0 & 1 & 0.5 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{[Cavalier projection onto XY]}} \cdot \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -5 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{[Move plane to origin } Z=0\text{]}} \cdot \underbrace{\begin{bmatrix} 10 \\ 20 \\ 40 \\ 1 \end{bmatrix}}_{\text{[Original Point P]}}$$

- What is vanishing point? Briefly explain.

2

All lines that are parallel to each other (in 3D space) appear to converge into a single point (in projected space). This point is called vanishing point.

3. Given COP at (-10, 90, 100), calculate the projected point P' for a point P (-245, 50, 30) where the Projection Plane (PP) is 200 units away from COP along X axis 10

$$X_p = -10 - 200 = -210 \text{ (along x axis)}$$

$$q = \text{COP} - (X_p, 0, 0) = (-10, 90, 100) - (-210, 0, 0) = (200, 90, 100) = (q_x, q_y, q_z)$$

next apply formulae (yz plane)

$$\begin{bmatrix} x' \cdot w \\ y' \cdot w \\ z' \cdot w \\ w \end{bmatrix} = \begin{bmatrix} \left(\frac{-x_p}{q_x} \right) & 0 & 0 & \left(x_p + \frac{x_p^2}{q_x} \right) \\ \left(\frac{-q_y}{q_x} \right) & 1 & 0 & \left(x_p \cdot \frac{q_y}{q_x} \right) \\ \left(\frac{-q_z}{q_x} \right) & 0 & 1 & \left(x_p \cdot \frac{q_z}{q_x} \right) \\ \left(\frac{-1}{q_x} \right) & 0 & 0 & \left(1 + \frac{x_p}{q_x} \right) \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$